

Instructions on the final examination

Task

The final examination is a written research proposal. You are free to choose a research topic that aligns with your interests. The methodology in your research proposal must incorporate computational tools and techniques. The proposal should be concise, with a maximum length of one page (excluding references and AI use acknowledgment).

Your proposal should include the following components: an introduction that presents the problem your research aims to solve and explains its significance through a brief review of the literature, clear research questions, a detailed methodology section incorporating computational tools and techniques, and a discussion of the limitations of the methods you have chosen.

Structure

It is recommended that you follow the structure below.

Introduction (1st paragraph): Introduce the problem your research aims to solve and explain its significance in the context of current knowledge in the field. This introduction paragraph should provide a brief literature review of the background information, highlighting the gap or challenge that your research will address. Be sure to include citations from reliable sources to support the relevance and importance of the problem.

Research questions (2nd paragraph): Present your research question(s), limiting them to no more than three. Each research question should be clearly articulated, and if applicable, you should also outline any hypotheses that you intend to test. These questions and hypotheses should directly relate to the problem introduced in the first paragraph.

Methodology (3rd paragraph): Describe the methods and data sources you will use to answer your research questions. Specify the computational tools and techniques you will employ, such as data collection methods, analysis techniques, and software applications. Detail each step of your approach.

Limitations (4th paragraph): Discuss the limitations of your chosen methods. Consider potential biases, data limitations, and the assumptions inherent in your computational methods. Explain how these limitations might affect the validity or reliability of your results and what steps you could take to mitigate these challenges.

References: List all references in [APA style \(7th edition\)](#). Every source cited in the text must appear in the list, and every entry in the list must be cited in the text. Make sure to include DOIs or URLs where available.

AI use acknowledgment: Your proposal must end with an AI use acknowledgment. In a short paragraph or table, state: 1) which AI tools you used (e.g., ChatGPT, Claude); 2) what you used them for (e.g., learning a method, searching the literature, language editing); and 3) which parts of the proposal were affected, and how you checked the output (e.g., introduction: verifying every reference against the original source; methodology: checking AI-suggested code against a package's documentation). If you did not use AI tools, state this instead.

Submission

The deadline for submission is February 19, 2027. Please email your proposal in PDF format to cai@verkehr.tu-darmstadt.de. Please make sure that the file is named according to the following format: MatriculationNumber_LastName_FirstName.pdf

Evaluation

Important: Please make sure that you adhere to the [TUDa guidelines on the use of AI](#). The text of the proposal must be written by you. Undeclared AI use or submitting fully AI-generated text as your own work is treated as an attempt to cheat, and the examination will be graded as a fail (5.0).

The following criteria are used for the evaluation of your research proposal.

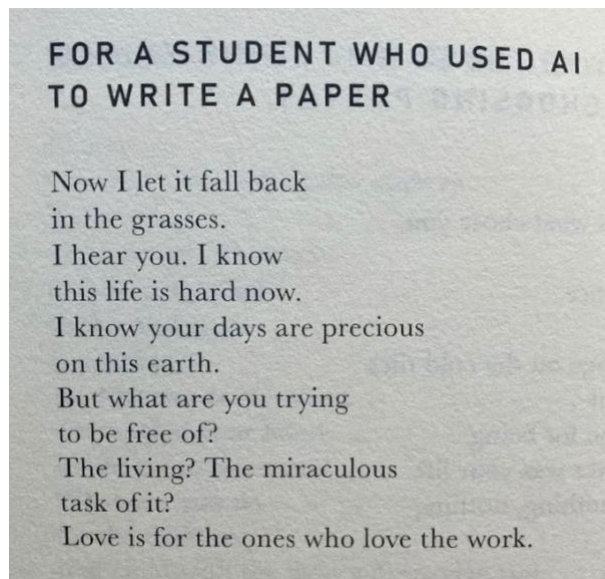
Introduction (20 points, 20%)	
Very Good (18–20 points)	Offers a logical argument supported by a coherent review of the literature. Demonstrates how existing studies inform the proposed research and uses all sources appropriately.
Good (15–17 points)	Presents a clear argument with some sources appropriately used but lacks depth or insights.
Satisfactory (12–14 points)	Provides a general literature review but lacks coherence and depth.
Fair (10–11 points)	Shows minimal engagement with the literature and makes few connections to the research questions.
Fail (0–9 points)	Little or no use of literature; logical flow is absent.
Research questions (20 points, 20%)	
Very Good (18–20 points)	Clearly states well-defined research question(s), appropriate in scope and relevance. Provides clear descriptions for non-expert audiences and adopts an appropriate academic tone.
Good (15–17 points)	Research question is defined but may need more precision or scope adjustment.

Satisfactory (12–14 points)	Research question is present but lacks clarity or depth.
Fair (10–11 points)	Purpose is implied, but the question is vague.
Fail (0–9 points)	Research question is missing or completely unclear.
Methodology (20 points, 20%)	
Very Good (18–20 points)	Clearly outlines methodology, incorporating computational methods. Describes the data sources and process in a way that can be replicated by others.
Good (15–17 points)	Includes a sound methodology but lacks some details about computational tools or replicability.
Satisfactory (12–14 points)	Methodology is described, but some steps are missing or unclear.
Fair (10–11 points)	Methodology is unclear or too vague.
Fail (0–9 points)	No clear methodology provided.
Limitations (20 points, 20%)	
Very Good (18–20 points)	Clearly identifies the study's limitations. The limitations are logical and well-argued.
Good (15–17 points)	Clearly identifies limitations but lacks depth.
Satisfactory (12–14 points)	Limitations are mentioned but are not well-argued.
Fair (10–11 points)	Limitations are mentioned but are illogical.
Fail (0–9 points)	Limitations are missing.
Writing (20 points, 20%)	
Very Good (18–20 points)	Writing is clear, concise, and well-organized with a logical flow. Minimal errors in grammar, punctuation, or spelling. All references are cited correctly in APA style. The AI use acknowledgment is complete and specific about tools, purposes, affected sections, and how outputs were checked.
Good (15–17 points)	Writing is mostly clear and organized, with few errors. The argument generally flows well, though some transitions may be weak. References contain minor formatting errors or inconsistencies. The AI use acknowledgment is complete but lacks some detail.
Satisfactory (12–14 points)	Writing is understandable but contains several errors. The structure is logical but may lack coherence in places. References contain several formatting errors or mismatches with in-text citations. The AI use acknowledgment is vague about purposes, affected sections, or how outputs were checked.
Fair (10–11 points)	Writing is often unclear with frequent errors. The structure is disjointed. References are inconsistently formatted or incomplete. The AI use acknowledgment is incomplete.
Fail (0–9 points)	Writing is unclear, difficult to follow, and lacks logical flow. References are largely missing. The AI use acknowledgment is missing.

The total points are converted to grades according to the table below.

Points	Grade
95–100	1.0 (very good)
90–94	1.3 (very good)
85–89	1.7 (good)
80–84	2.0 (good)
75–79	2.3 (good)
70–74	2.7 (satisfactory)
65–69	3.0 (satisfactory)
60–64	3.3 (satisfactory)
55–59	3.7 (sufficient)
50–54	4.0 (sufficient)
0–49	5.0 (fail)

Closing words



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